

Experiment No: 2 Exporatory Data Analysis(import and export files)

Aim:

To import and export different types of files using python.

Algorithm:

1. **Start** the program.
2. **Import** the required library (pandas).
3. **Specify** the path or name of the CSV file.
4. **Read** the CSV file using the read_csv() function.
5. **Store** the imported data in a DataFrame (df).
6. **Display** the first few records using head().
7. **Check** the dataset information using info().
8. **Display** the statistical summary using describe().
9. **Verify** that the dataset has been imported successfully.
10. **Stop** the program.

Program:

Importing csv files:

```
import pandas as pd
df=pd.read_csv(r"C:\Users\SHREE HARINI\Downloads\social_media_dataset_with_null_duplicates
print(df)
```

	Post_ID	Platform	Likes	Comments	Shares	Views	Sentiment
0	1	Instagram	304.0	384.0	71	17049	Positive
1	2	Facebook	939.0	351.0	190	36741	Positive
2	3	YouTube	250000.0	21.0	8	7140	Positive
3	4	Facebook	4239.0	313.0	7	37781	Positive
4	5	YouTube	3536.0	117.0	115	39618	Neutral
5	6	Instagram	NaN	362.0	109	23298	Neutral
6	7	Facebook	1863.0	495.0	196	23059	Positive
7	8	Instagram	3212.0	54.0	92	23541	Negative
8	9	Twitter	455.0	378.0	118	36142	Positive
9	10	LinkedIn	745.0	287.0	76	42198	Negative
10	11	Twitter	4829.0	103.0	181	5558	Positive
11	12	Facebook	2470.0	45.0	60	7619	Neutral
12	13	NaN	3814.0	330.0	94	11659	Neutral
13	14	Twitter	1816.0	348.0	69	46994	Negative
14	15	Instagram	1501.0	278.0	187	17043	Positive
15	16	LinkedIn	3208.0	143.0	164	46099	Negative
16	17	Facebook	2756.0	436.0	197	4665	Positive
17	18	Instagram	2684.0	210.0	69	5337	Positive
18	19	YouTube	2677.0	113.0	168	1000000	Neutral
19	20	LinkedIn	1270.0	140.0	36	17162	Negative
20	21	YouTube	4515.0	NaN	192	39311	Neutral
21	22	YouTube	3371.0	190.0	57	10065	Negative
22	23	LinkedIn	844.0	391.0	13	8185	Positive
23	24	Facebook	3558.0	310.0	17	26216	Neutral
24	25	YouTube	3934.0	275.0	65	37256	Positive
25	26	Instagram	4498.0	389.0	69	43006	Neutral
26	27	Instagram	2504.0	227.0	41	30735	Positive
27	28	Twitter	4200.0	395.0	46	34271	Positive
28	29	Twitter	4258.0	316.0	51	11016	Neutral

```
print(df.head())
```

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```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 53 entries, 0 to 52
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Post_ID     53 non-null    int64
1   Platform    52 non-null    object
2   Likes       52 non-null    float64
3   Comments    52 non-null    float64
4   Shares      53 non-null    int64
5   Views       53 non-null    int64
6   Sentiment   52 non-null    object
dtypes: float64(2), int64(3), object(2)
memory usage: 3.0+ KB
None
```

```
print(df.describe())
```

	Post_ID	Likes	Comments	Shares	Views
count	53.000000	52.000000	52.000000	53.000000	53.000000
mean	25.018868	7544.961538	345.115385	96.584906	45828.415094
std	14.386814	34306.916476	674.785214	61.786367	134280.320211
min	1.000000	220.000000	8.000000	5.000000	4665.000000
25%	13.000000	1893.750000	116.000000	51.000000	16491.000000
50%	25.000000	3038.500000	276.500000	76.000000	31318.000000
75%	37.000000	3978.750000	385.250000	151.000000	38182.000000
max	50.000000	25000.000000	5000.000000	200.000000	100000.000000

Exporting csv file:

```
df.to_csv('output.csv',index=False)
print("CSV file Exported")
```

CSV file Exported

Visualization:

```
import matplotlib.pyplot as plt
import seaborn as sns

platform = df["Platform"].value_counts()

plt.figure(figsize=(6,6))

plt.pie(
    platform,
    labels=platform.index,
    autopct="%1.1f%%"
)

plt.title("Platform Distribution")

plt.show()
```

Platform Distribution

